

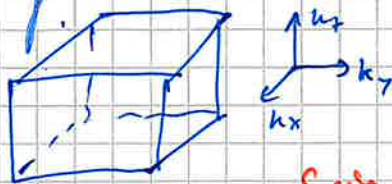
2 Quantum dots and impurities

2.1 Electron gases in 3, 2, 1 and 0(!) dimensions

3D electron gases: metals and semiconductors

non-interacting electrons on periodic lattices:

Bloch's theorem explains why electrons move (almost) like free electrons



free electron wave function:

$$\psi_{\vec{k}\sigma}(\vec{r}) = \frac{1}{\sqrt{V}} e^{ik_x x} e^{ik_y y} e^{ik_z z} \chi_{\sigma}$$

Spin part

free propagation in all directions

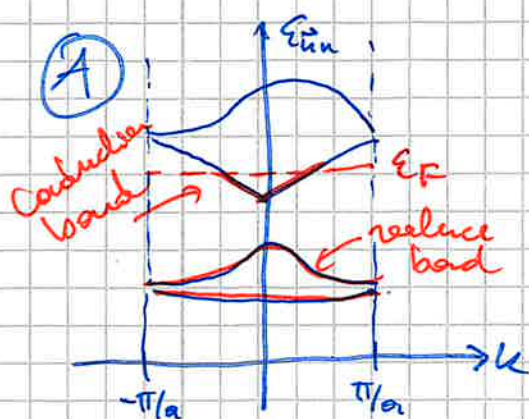
$$E_{k_x, k_y, k_z} = \frac{\hbar^2}{2m^*} (k_x^2 + k_y^2 + k_z^2)$$

modification due to lattice
→ effective mass

allowed energies in solid state depend on crystal structure

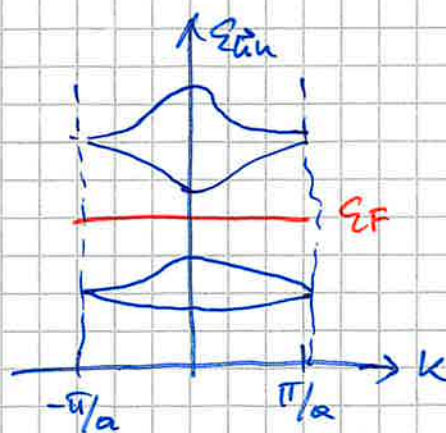
→ $E_{\vec{k}n}$... band structure / dispersion
band index

two principle cases are possible



Fermi level lies within band

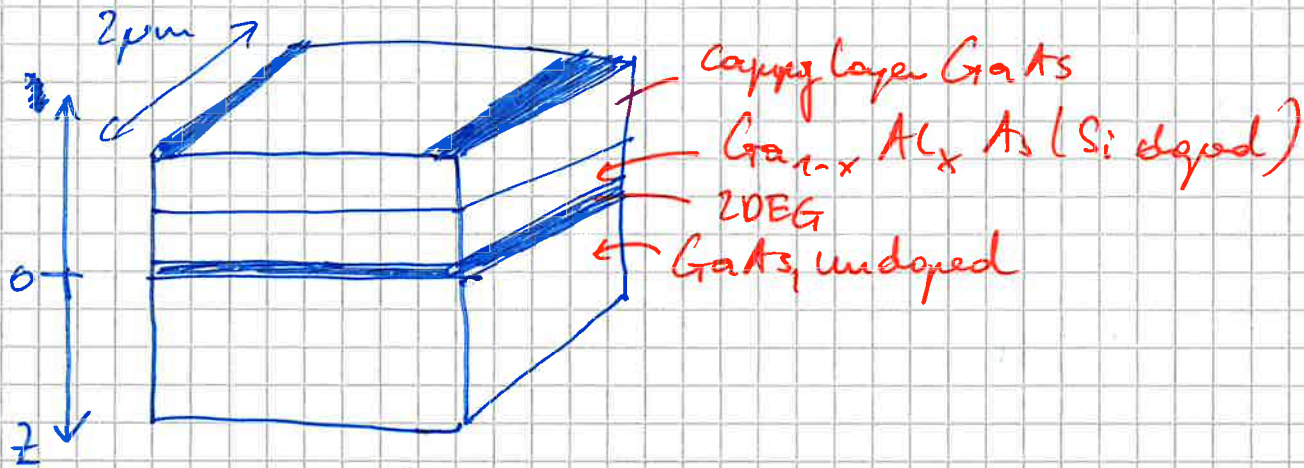
⇒ metal



Fermi level lies in gap between
conduction and valence band

⇒ insulator / semiconductor

2D electron gases: GaAs/GaAlAs heterostructures

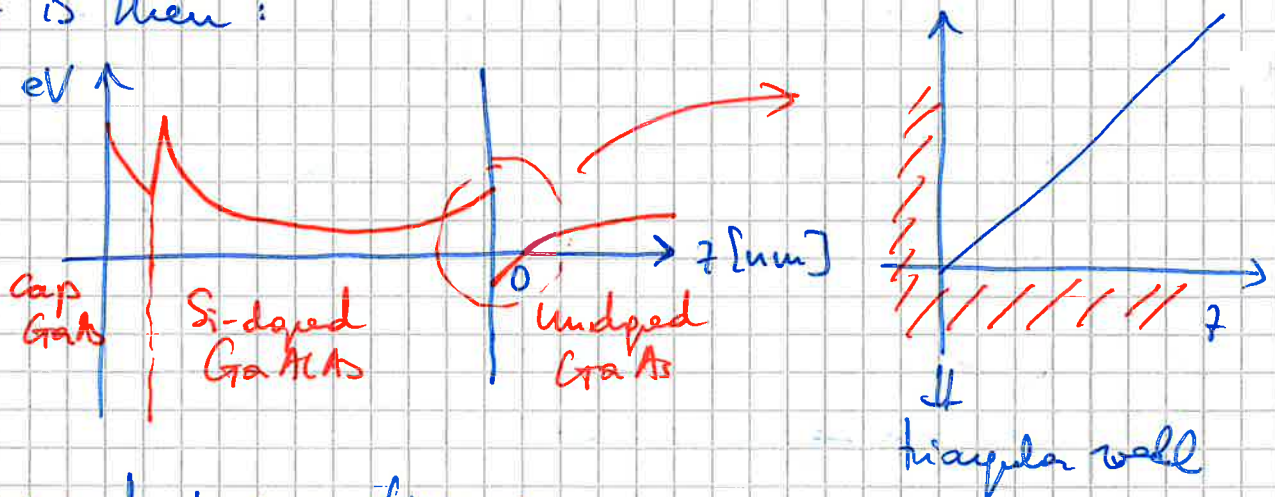


2DEG... two-dimensional electron gas

These heterostructures can be grown with more atomic precision by molecular beam epitaxy (MBE)

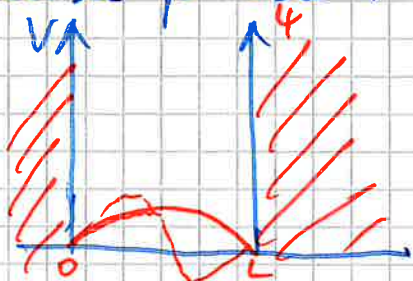
GaAs / GaAlAs are both semiconductors, however, the conduction band for $\text{Ga}_{1-x}\text{Al}_x\text{As}$ lies 300 meV higher than GaAs

The conduction band for the heterostructure in dependence of z is then:



Confined in one dimension

particle in a box:



Schrodinger equation $\hat{H}\psi = E_n \psi$
with $E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$
 $\Rightarrow$ quantized energies!

here: $\psi_{k_x, k_y, n}(\vec{r}) = \frac{1}{\sqrt{A}} e^{ik_x x} e^{ik_y y} \varphi_n(z) \chi_\sigma$

free propagator in x, y

in the extension of hetero well

$$\epsilon_{k_x, k_y, n} = \frac{\hbar^2}{2m^*} (k_x^2 + k_y^2) + \epsilon_n$$

examples (2DEG): quantum Hall effect, lasers, semiconductor heterostructures

A similar effect occurs in 1D: Carbon nanotubes

There, $\psi_{n, e}(\vec{r}) = \frac{1}{\sqrt{L}} e^{ik_x x} R_{n, e}(r) Y_e(\phi) \chi_\sigma$

cylindrical wave functions

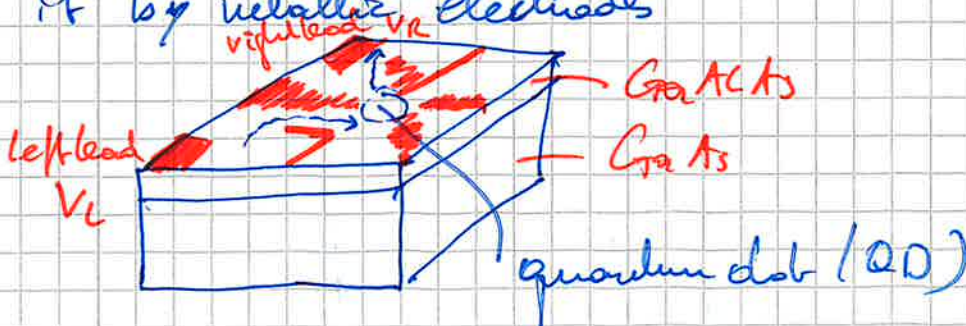
free only in x -direction

$$\epsilon_{n, n, e} = E_0^R + E_n^\phi + \frac{\hbar^2}{2m} k_x^2$$

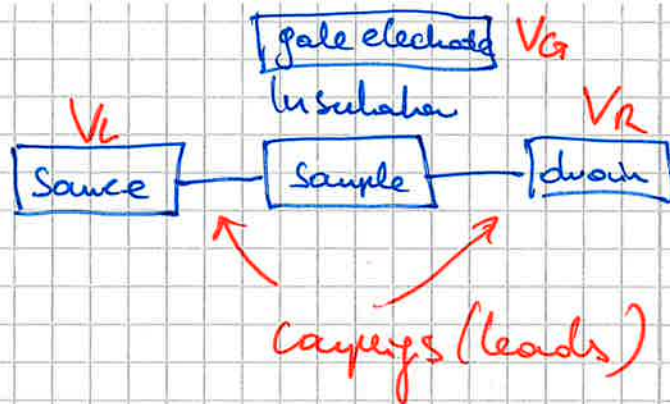
0D electron gases: quantum dots / artificial atoms

If we confine the electron gas in all directions, we only have discrete energy levels $\Rightarrow$ "atom" at low T ($\sim 100 \text{ mK}$)

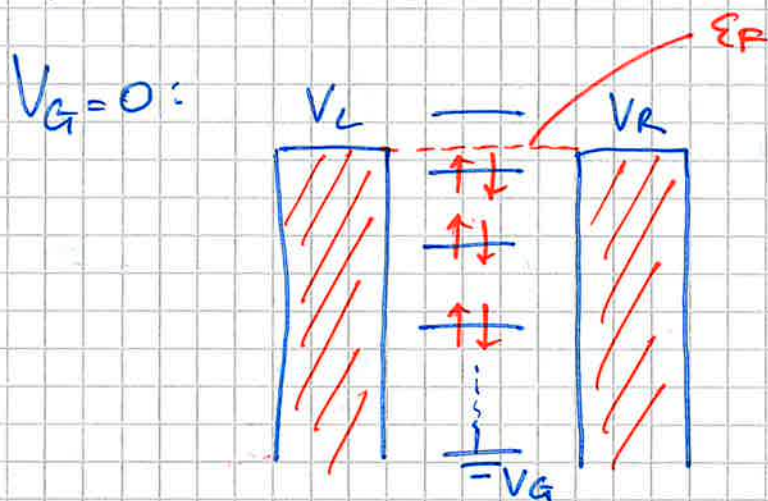
This can be done by growing a 2DEG and confining it by metallic electrodes



Simple depiction:

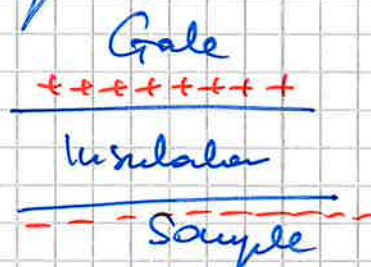


at weak coupling, the QD only receives "single" electrons from Source. The "atomic" energy levels are filled up ~~to~~ the Fermi level. let's consider $V_G = 0$ first:



What is the influence of the gate voltage V_G ?

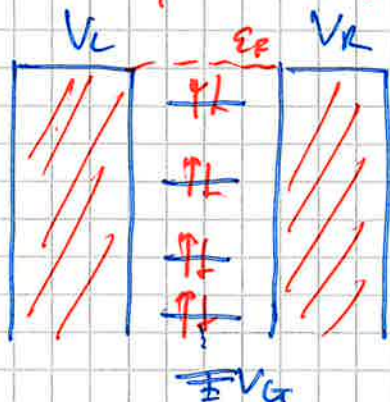
Consider again the sample model:



$\Rightarrow$ the device acts as a capacitance with capacity C

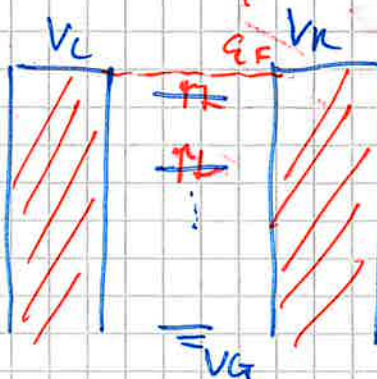
$V_G = +V$

$\Rightarrow$ adds e^- , raises E_F



$V_G = -V$

$\Rightarrow$ removes e^- , ~~lowers~~ lowers E_F



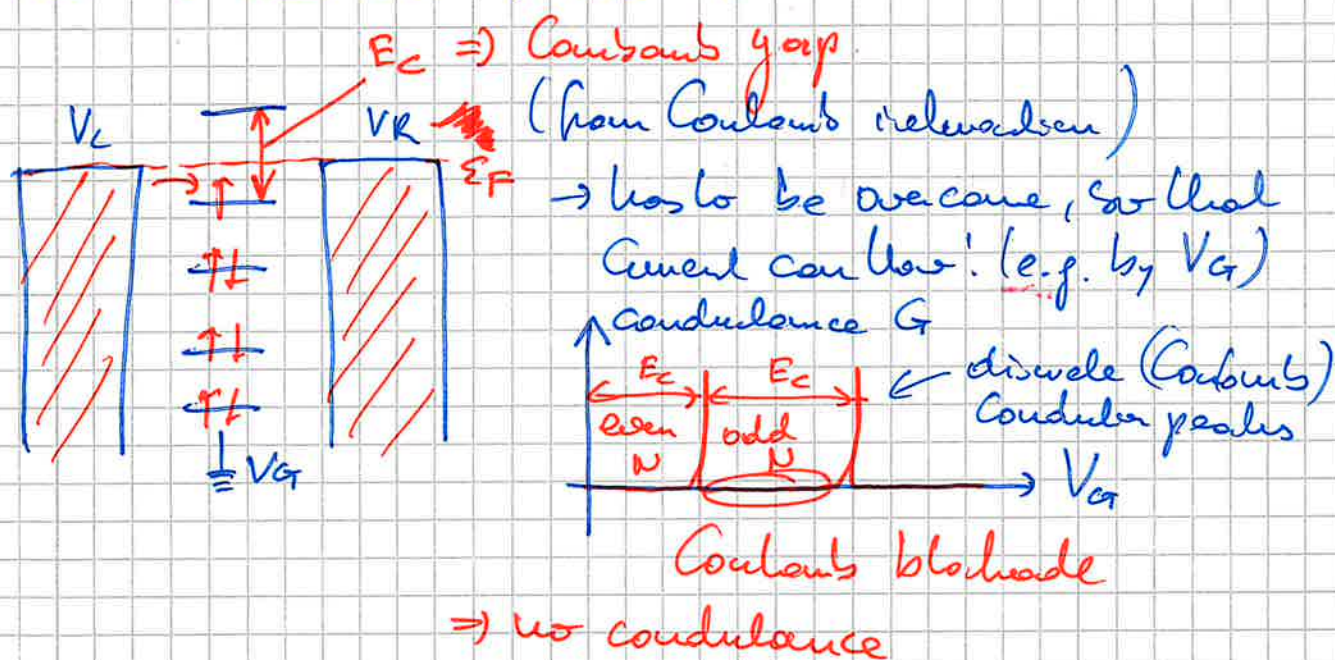
Conductance through quantum dot

adding an electron changes the energy of a QD

→ constant-interaction model: Coulomb interaction between electrons is constant w.r.t. number of e^-

→ electrostatic charging energy $E_c = \frac{Q^2}{2C}$ ^{charge} _{capacitance}

For a current to pass, an electron has to tunnel to the QD and out of the QD



Conduction only occurs if $E_N \neq E_{N+1}$

$$E_N = \underbrace{\epsilon_1 + \epsilon_2 + \dots + \epsilon_N}_{\text{energy levels filled}} + \underbrace{\frac{(Ne)^2}{2C}}_{\text{charging energy}}$$

$$E_{N+1} = \epsilon_1 + \dots + \epsilon_{N+1} + \frac{(N+1)^2 e^2}{2C}$$

$$E_{N-1} = \epsilon_1 + \dots + \epsilon_{N-1} + \frac{(N-1)^2 e^2}{2C}$$

energy differences:

$$\Delta E_{N \rightarrow N+1} = \varepsilon_{N+1} + (2N+1) \frac{e^2}{2C}$$

$$\Delta E_{N-1 \rightarrow N} = \varepsilon_N + (2N-1) \frac{e^2}{2C}$$

Coulomb diamonds occur if $E_N = E_{N+1}$ or $E_{N-1} = E_N$

$\Rightarrow$ Separation of Coulomb diamond peaks:

$$\Delta E_{N \rightarrow N+1} - \Delta E_{N-1 \rightarrow N} = \underbrace{\varepsilon_{N+1} - \varepsilon_N}_{\Delta \varepsilon \text{ level spacing}} + \underbrace{\frac{e^2}{2C}}_{\text{charging energy}}$$

